



TF2AngleNet: Continuous finger joint angle estimation based on multidimensional time–frequency features of sEMG signals

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ABSTRACT

Current pattern recognition-based myoelectric prosthetic hand control methods map electromyography (EMG) signals to specific hand postures, achieving high accuracy but often resulting in unnatural movements during transitions, reducing the hand's anthropomorphic nature. While some studies predict single-finger joint angles from EMG signals, these approaches lack practicality since arm muscles often control multiple fingers simultaneously. This study proposed a TF2AngleNet that predicts six finger joint angles using both time domain raw signals and frequency domain features of EMG signals. A novel non-contact joint angle measurement method was used to collect EMG and joint angle data from five healthy subjects over five days. The experimental results demonstrate that TF2AngleNet achieves outstanding performance in continuous joint angle estimation, with a correlation coefficient of 94.7%, an R^2 value of 89.2%, and an NRMSE of 9.5%. Notably, this represents a 12.43% improvement in NRMSE, along with average gains of 1.2% in CC and 2.42% in R^2 compared to single-domain models (p-values < 0.05 across all metrics). Also, hand postures were shown using a virtual hand model, providing a natural and bionic control method of myoelectric hands. Additionally, a novel conceptual framework is proposed to reduce barriers to using pattern recognition-based prosthetic hands, with this study serving as its first stage by validating the model's performance under three experimental conditions. This research provides a promising solution for dexterous, biomimetic and practical myoelectric prosthetic hand control methods.

1. Introduction

Around the world, there are numerous individuals who have lost their arms due to accidents or other reasons. To restore amputees' lost hand function, prosthetic hands are made more and more dexterous and with more degrees-of-freedom (DOF) [1–3]. Consequently, the market offers various types of prosthetic hands, among which multi-DOF myoelectric prosthetic hands based on surface electromyography (sEMG) signals showed their effectiveness [2]. This is because sEMG could be acquired in a non-invasive way by using only a few pairs of sEMG electrodes attached to users' forearm skin. Therefore, this research focuses on using sEMG signals to predict users' hand motion.

However, EMG signal is a kind of fragile electrical signal generated by muscle activity and is susceptible to alteration by factors such as

sweat, muscle fatigue, and electrode displacement [4,5]. Meanwhile, EMG signals can vary greatly between healthy individuals due to each individual's muscle conditions such as fat thickness and muscle strength [6]. These factors caused commercial prosthetic hands to choose control methods that utilizes robust features of sEMG signal such as the amplitude of sEMG signal.

The proportional control method and threshold control method are representative control strategies for commercial prosthetic hands, such as the Myobock (Ottobock, Germany). In threshold control methods, the open and grasp motions are controlled by a simple mechanism that compares the real-time amplitude value of sEMG signals with a preset threshold value. For example, the robotic hand would do grasp motions when the real-time amplitude value exceeds the preset value.

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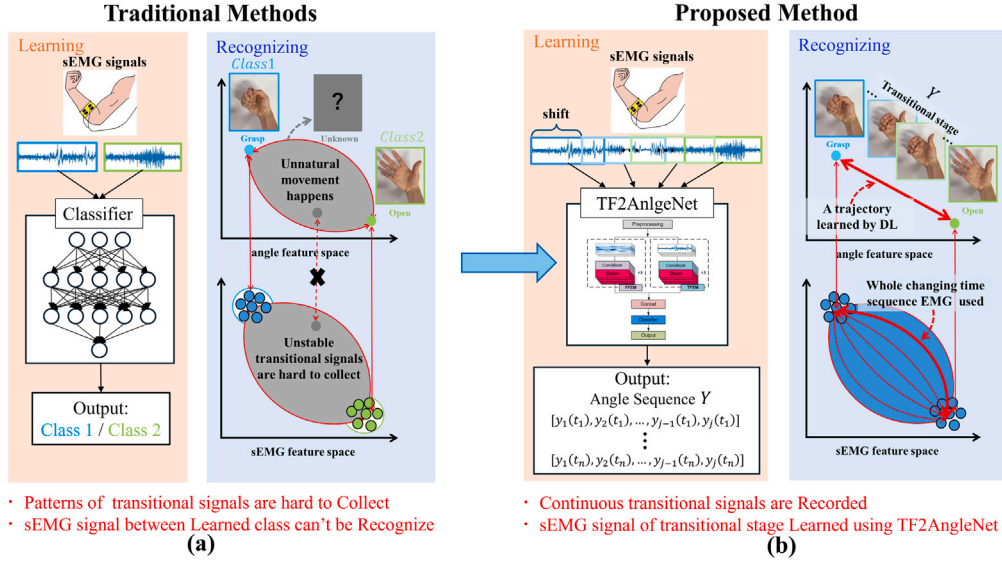


Fig. 1. Concept comparison between traditional methods and the proposed method. DL: Deep learning methods. j : The number of angles for prediction. n : The length of the joint angle sequence Y .

Proportional control methods map the amplitude value to the grasping force or grasping angle of the hand. These control strategies are robust but cannot achieve diverse motions because of the insufficient effective information from the amplitude value. This makes that only prosthetic hands with low-DOF adopt the strategies above. To control dexterous prosthetic hands with higher DOF, switching control methods are widely adopted, such as the Bebionic Hand (OttoBock, Germany). This kind of method uses a button or a certain co-activation pattern of sEMG signals to switch from the current posture to the desired posture. Then, the hand is flexed and opened by using proportional control or threshold control. This kind of method achieves diverse motions by using an unnatural way to change from one hand posture to another, sacrificing the intuitiveness of the prosthetic hand.

Moreover, with recent advancements in machine learning methods and artificial neural network algorithms, pattern recognition-based methods have been demonstrated to be not only useful in industrial fields but also in medical applications [7–9]. Especially in the area of prosthetic hand control, pattern recognition-based control methods have been demonstrated to be highly accurate in predicting users' motion intentions [10–13]. This kind of control approach involves measuring the myoelectric signals generated by the contraction of their residual muscles and training a machine learning algorithm to map these signals to corresponding motions, ultimately enabling the intuitive operation of the prosthetic hand for specific actions. However, most of the current pattern recognition-based methods simply pair different movements with different EMG patterns, leading the robotic hand to generate discrete outputs. Unnatural movements happens when the pattern recognition-based prosthetic hand tries to switch one posture to another. The aim of this research is to develop a control algorithm that maps sEMG signals with continuous joint angles of the hand, enabling the hand's movement more biomimetic.

Continuous joint angle estimation has been researched widely in recent years. Pan et al. [14] investigated continuous estimation of individual finger's two joint angles under different wrist motions. Long et al. [15] developed a method based on deep learning to predict 10 joint angles of five fingers' metacarpophalangeal (MCP) and proximal inter-phalangeal joints. Some previous research focus on predicting the continuous movement of hand's main MCP joints using sEMG signals [16–20].

In these previous research, either the movement of individual fingers was chosen as learning data for the algorithm, or the coordinated movement of fingers was randomly selected. In this situation, the

motion mechanism of the human hand was not considered because single forearm muscle is physically connected to the motion of multiple fingers [21,22]. To address this gap in the research field of multi-joint coordinated control, this study leverages the theory of postural synergies to select practical target movements for prosthetic hand control [23]. However, in traditional deep neural networks, a higher number of output neurons leads to a greater number of learnable parameters, which diminishes the practical applicability of the algorithm. To mitigate the increase in learnable parameters caused by the expansion of control objectives, a convolutional neural network is employed to design a more practical control algorithm. Therefore, this study seeks to integrate the mechanism of hand movement with a convolutional neural network to design a control method that is both more biomimetic and practical.

The objective of this study is to build a practical control algorithm for continuously controlling dexterous myoelectric prosthetic hands. Fig. 1(b) shows the concept of the proposed method: a regression model that is able to continuously predict hand postures based on sequential sEMG signals. Inspired by former convolutional neural network based algorithm [10,24], a TF2AngleNet was proposed to predict six finger joint angles using both time-domain raw signals and frequency-domain features of EMG signals. Several continuous hand movements based on postural synergies were chosen as teacher data, building a more natural mapping function from sEMG signals to continuously changing hand motion.

The main contribution of this research is as follows:

- A novel neural network model, named TF2AngleNet, is proposed for the continuous and simultaneous multi-DOF control of myoelectric prosthetic hands. This model integrates multi-dimensional time-domain and frequency-domain information, allowing the prediction of sequential continuous joint angle changes based on temporal sEMG signals. Experimental results demonstrated that the proposed model outperforms traditional models in enhancing the accuracy of motion intent recognition.
- This study introduced an innovative data collection system that combines non-contact finger joint angle measurement with traditional surface EMG signal acquisition. Unlike most previous methods that utilize data gloves for joint angle measurement, the proposed approach employs the Leap Motion Controller, enabling data measurement without direct sensor contact with the hand.

- A two-stage conceptual framework was proposed to address the high cost of data collection in controlling pattern-recognition-based myoelectric prosthetic hands. In the first stage, a large dataset from healthy individuals is used to pre-train a neural network to extract common EMG features. In the second stage, the user collects a small amount of data, and through a transfer learning approach, the pre-trained model from the first stage can be adapted, significantly reducing the data collection requirements.

The rest of this article is organized as follows: Section 2 details the proposed method, including data preprocessing and the development of TF2AngleNet. Section 3 outlines the experimental setup and the construction of the TF2AngleNet training data. Section 4 presents and discusses the experimental results, and Section 5 concludes the article.

2. Materials and methods

2.1. Design philosophy of the proposed approach

In traditional hand posture recognition algorithms, users perform different postures and maintain the muscle activity associated with each posture. The sEMG signals recorded during this period are used to train classification algorithms. As shown in Fig. 1(a), the EMG signals during the hand posture transition process cannot be recognized by the classification model because these signals are usually unstable, making it challenging to collect and classify such continuously changing data [25,26]. In general hand posture classification, the process of transitioning from one posture to another is a discrete changing process since any posture currently is defined as a combination of fixed positions of all active motors of the hand. Therefore, to construct an algorithm capable of predicting the joint angles of a hand during hand posture transitions, the continuously changing EMG features should be matched with the continuously changing active joint angles of the hand.

In this study, as shown in Fig. 1(b), a TF2AngleNet was proposed to perform regression prediction of the continuous changes in the six joint angles of prosthetic hands based on five-channel surface sEMG signals. As shown in Fig. 1(a), due to the instability of transition stages of sEMG signals, different signals could be mapped to a same combination of joint angles. This means the sEMG features' moving trajectory from a hand posture to another is unstable and diverse, requiring more data to stabilize a regression model. Recent studies have shown that deep artificial neural networks are superior to traditional machine learning algorithms in dealing tasks that require large data [27]. To solve this problem, extensive overlapped sEMG signals and angle variation data were used to train a deep artificial neural network. Therefore, the mapping function between sEMG signals and active joint angles during different hand posture transitions could be established.

2.2. The conceptual framework of this research

A preliminary research [10] has demonstrated that collecting multi-day data from the same user can alleviate the decline in prediction accuracy caused by sensor re-wearing. However, collecting multi-day data may lead users to abandon the use of pattern recognition-based myoelectric hands [28,29] due to the significant burden it imposes. To solve the above issue, this research proposed a two-stage conceptual framework. This research represents the first stage, building a network trained on a large, cross-day and cross-subject dataset to recognize common features in average and able-bodied people. As for the second stage, by developing and using a transfer learning method, a new user could achieve high-accuracy multi-motion recognition with a small amount of data by leveraging the trained network.

2.3. The proposed approach

As shown in Fig. 2, the proposed approach mainly comprises three parts: sEMG signal measurement and signal preprocessing, non-contact hand motion data measurement and data preprocessing, and finger joint angle prediction model.

2.3.1. sEMG signal measurement and signal preprocessing

The sEMG signals were acquired using custom-made sEMG sensors [30]. The sensor utilizes a differential amplifier to reduce noise in the sEMG signals and a band-pass filter to capture signals in the 10–400 Hz frequency range. The primary source of noise in sEMG signals is power line interference. Therefore, a notch filter was employed to remove the 50 Hz frequency band signal. Additionally, another amplifier was used to amplify the signal to approximately 80,000 times its original magnitude. An ad converter device was utilized to acquire 16-bit signals ranging from –10V to 10 V, which were then transmitted to a computer. The sampling frequency was set at 1600 Hz, as the subsequent sEMG signal preprocessing involved using short-time Fourier transform (STFT), which requires a sampling rate four times higher than the maximum signal frequency.

The measured sEMG signals were transformed into time-frequency domain feature maps using STFT [10]. Here, s_j denotes the signal sequence from channel j , and the spectrogram $f_t^k(s_j)$ at time t and frequency k can be defined as follows:

$$f_t^k(s_j) = \frac{1}{2I_p + 1} \sum_{n=k-I_p}^{k+I_p} |F(s_j, t, n)|^2 \quad (1)$$

$$F(s_j, t, n) = \sum_{m=1}^{N_{FL}+m} s_j(t - N_{FL} + m) \cdot e^{-i \frac{2\pi(m-1)n}{N_{FL}}} \quad (2)$$

$$ch : \{j \in \mathbb{N} \mid 1 \leq j \leq N_{ch}\}$$

$$time : \{t \in \mathbb{N} \mid 1 \leq t \leq N_{time}\}$$

$$: \{m \in \mathbb{N} \mid 1 \leq m \leq N_{FL}\}$$

$$freq : \{k \in \mathbb{N} \mid 1 \leq k \leq N_{spt}\}$$

$$k : \{n \in \mathbb{N} \mid k - I_p \leq n \leq k + I_p\}$$

where N_{ch} denotes the channel size; N_{FL} indicates the length of the frame; N_{time} and N_{spt} represent the numbers of samples required to extract the time and frequency, respectively; I_p denotes the smoothing width; and $F(s_j, t, n)$ represents the Fourier transform of the signal sequence from channel j , calculated using the data from the current time t going backward for a length of N_{FL} , capturing the signal's frequency components at frequency index n . Table 1 shows the actual parameters for generating spectrograms comprised of time–frequency domain features. After performing STFT, a one-second sEMG sequence s_j from channel j is transformed into a 16×1089 time–frequency spectrogram, where 16 corresponds to the number of frequency components and 1089 represents the number of time points. Considering the high real-time requirements of sEMG signal recognition and the limited computational resources, we performed temporal averaging on the 16×1089 spectrogram along the time dimension, reducing it to a 16×100 feature map. The spectrograms were extracted while the time was shifted (shift size = 500) to include each other in the range, which stabilized the recognition of the sEMG signals.

2.3.2. Non-contact hand motion data measurement strategy and joint angle preprocessing

The coordinate information of all 24 finger joints was measured using a Leap Motion Controller 2 (Leap Motion, Inc., USA). To apply this control algorithm to a myoelectric prosthetic hand under development, which has six active DOF (thumb flexion/extension and rotation, as well as MCP joints of the index, middle, ring, and little fingers), the measured coordinate information was utilized to calculate continuously varying data for the six joint angles [31]. All joint angles were computed using the dot product between two vectors, which represent the segments adjacent to the joint. Subsequently, the range of the translated joint angles between 0 and 1 was scaled to normalize and accelerate the proposed neural network training process; herein, the values of 0 and 1 represented maximal extension and flexion of fingers, respectively. To build the mapping function between sEMG signals and joint angles, the outputs corresponding to the spectrograms should not be individual

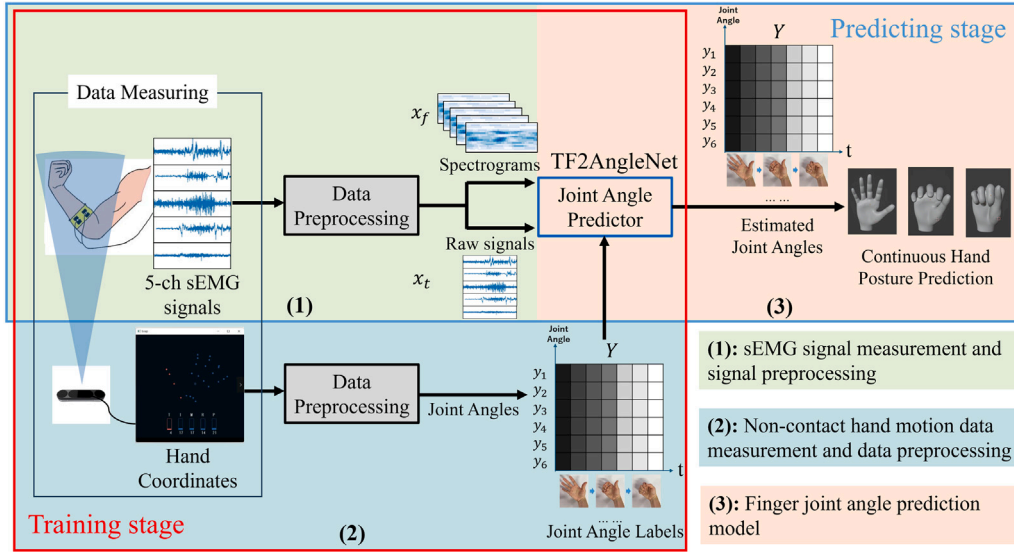


Fig. 2. Proposed approach.

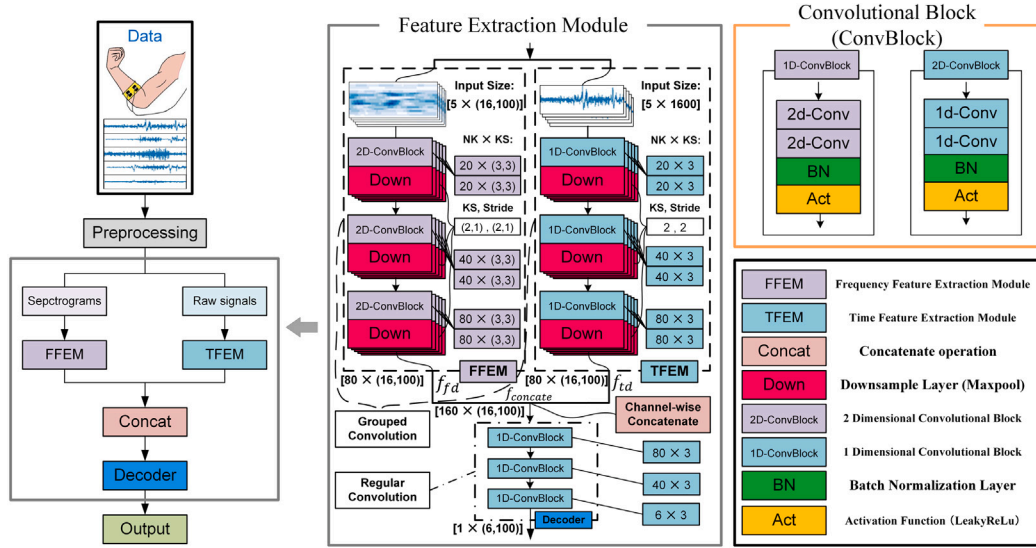
Fig. 3. Network structure proposed in the study. NK: Number of kernels; KS: Kernel size; f_{fd} and f_{td} : Feature maps from the frequency domain and time domain, respectively; f_{concat} : Concatenated feature maps.

Table 1
Actual parameters for spectrogram generation.

I_p	N_{FL}	N_{ch}	N_{time}	N_{spect}
2	512	5	1600 (= 1 s)	16 (= 400 Hz)

predictions of joint angles but their continuous changes. Although the hand motion sampling frequency was the same as sEMG signal sampling (1600 Hz), the effective sampling rate was set as 50 Hz. To smooth the prediction, the prediction value was set to 100 Hz, resulting in each input data corresponding to 6×100 joint angle values.

2.3.3. Finger joint angle prediction model

sEMG signals are inherently fragile and susceptible to noise interference. Therefore, an robust feature engineering method is important for extracting motion intention from users. Some previous studies have shown that combining frequency domain and time domain features of sEMG signals is helpful to improve the accuracy of predicting users' motion intention [32,33]. In this study, a method that extracts and

integrates features from both the frequency and time domains of the same sEMG signals was employed, which possess different dimensions. As shown in Fig. 3, preprocessed sEMG signals can be primarily divided into two types of features: frequency domain features extracted in Section 2.3.1. and raw sEMG signals used as time domain features.

Recent research indicates that deep artificial neural networks have advantages over traditional machine learning methods in model learning from large datasets. Given that this study aims to address the task of continuous variation of joint angles from sEMG signals, which involves learning the mapping relationship between high-dimensional input and output spaces, traditional neural network structures such as multi-layer perceptrons result in a vast number of parameters, limiting the practicality of control methods. The development of convolutional neural networks (CNN) in recent years has demonstrated the potential to solve such tasks, with advanced neural networks achieving high-precision pixel-level predictions. Therefore, this study adopts a TF2AngleNet capable of integrating different dimensional features from sEMG signals for precise regression prediction of hand postures, as illustrated in Fig. 3. The continuous joint angle prediction network structure used in this

study can be represented by the following decision function:

$$Y_{pred} = F_w(x_f, x_t) \quad (3)$$

where x_f and x_t represent the two-dimensional (2D) spectrograms and one-dimensional (1D) raw sEMG data extracted from a one-second window, respectively. F_w is the decision function, w represents all learnable parameters of the network, and Y_{pred} denotes the predicted 2D joint angle sequence. By learning this decision function, the proposed network can predict multiple angle sequences from the signal sequence. It is expected to use features from both the time and frequency domains to stabilize these predictions.

As shown in Fig. 3, after data preprocessing, the sEMG signals pass through a two-stream feature extraction module designed to simultaneously extract 2D frequency domain features and 1D time domain features. For the 2D time-frequency domain features, three 2D convolutional blocks consisting of two 2D convolutional layers, a batch normalization layer, and an activation function layer were utilized to extract frequency domain information, as illustrated in Fig. 3. For the 1D features, feature extraction is performed using three 1D convolutional blocks with a similar structure. Each 1D or 2D convolutional block is designed with two convolutional layers using the same number of kernels. The number of convolutional kernels in the first, second, and third 1D/2D convolutional blocks is set to 20, 40, and 80, respectively. The gradually increasing number of convolutional kernels is expected to facilitate the extraction of more high-dimensional features. For 2D and 1D features, convolutional kernels of size (3, 3) and length 3, respectively, are employed. Maxpool is applied as the downsampling layer, with the filter size and stride set to (2, 1) for 2D features and 2 for 1D features. Since each channel of the sEMG signals represents muscle activity of different muscles, it is necessary to extract features for each channel separately. Therefore, for both types of features, a grouped convolution strategy was employed, which maximizes the utilization of each channel's features while maintaining model lightweight characteristics.

The layer for concatenating different domain features is defined as follows:

$$f_{concat} = [f_{fd}^{(1)}, f_{fd}^{(2)}, \dots, f_{fd}^{(c)}, f_{td}^{(1)}, f_{td}^{(2)}, \dots, f_{td}^{(c)}] \quad (4)$$

where f_{fd} and f_{td} represents the feature maps of the c th channel from the frequency domain and time domain, respectively, and c is the total number of channels. The concatenation operation combines these feature maps along the channel dimension, resulting in a new feature map f_{concat} with $2c$ channels.

In traditional CNNs for hand posture estimation [10,11,24], a feature extractor composed of convolutional layers is followed by fully connected layers for classification. However, for the high-dimensional sequence-to-sequence prediction problem in this study, fully connected layers would introduce excessive parameters. To address this, a decoder with three 1D convolutional blocks was designed, allowing efficient feature interaction across channels while maintaining a lightweight model. The convolutional block structure used in the regression layers is the same as that employed earlier. For the regression layers, three 1D convolutional blocks were employed, with the number of convolutional kernels set to 80, 40, and 6 for the first, second, and third blocks, respectively. All convolutional kernels in the decoder had a length of 3. These parameters regarding the model architecture were primarily determined through iterative tuning during the training process, considering model performance, lightweight design, and empirical guidelines.

3. Experiment

In this section, a detailed description of the experimental conditions and the evaluation metrics used to assess the experiment is provided. All experimental procedures were approved by the Ethics Committee of The University of Electro-Communications, Tokyo, Japan

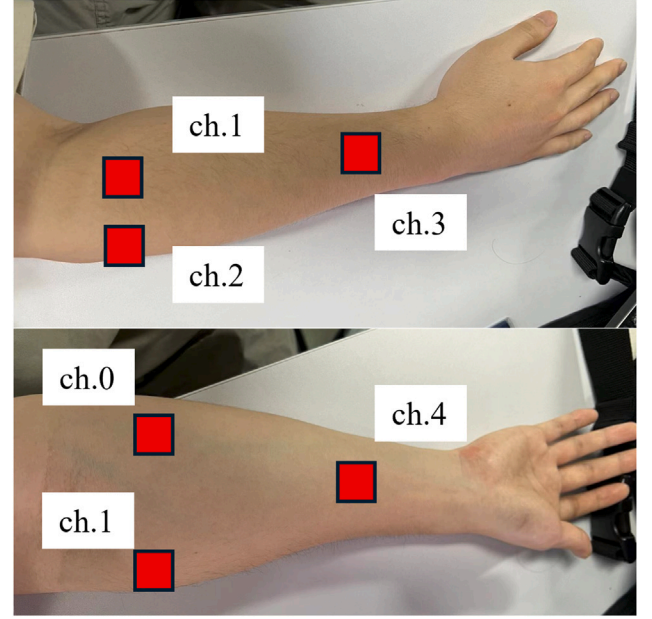


Fig. 4. sEMG measurement locations.

(Approval Number: H10006(7), Date of Approval: June 22, 2023). The experimental conditions will first describe the measurement sites for sEMG signals, the equipment used for measuring sEMG signals and finger joint angles, and the software employed for data collection. Subsequently, the method for constructing the dataset will be outlined. In the introduction of the evaluation metrics, their calculation methods and the rationale behind their use are described.

3.1. Experimental setup

3.1.1. Sensor attachment

To establish a mapping relationship between sEMG signals and the continuous variation of each finger joint angle, it is essential to ensure that the collected sEMG signals encompass most of the finger activity information. To achieve this objective, five sEMG sensors were attached near the following muscles: flexor digitorum superficialis, extensor digitorum, abductor pollicis longus, extensor indicis, and extensor digiti minimi. Fig. 4 illustrates the attachment locations of the sEMG sensors, which were secured on two elastic bands, with the positioning informed by prior studies [10,11] to ensure optimal signal acquisition and consistency with established methodologies. Fig. 5 depicts the scene of measuring data. Considering the variability in arm thickness among participants, the exact placement of sensors was determined through external palpation for each participant. Therefore, the sensor positions were adjusted prior to measuring the sEMG data for each participant. The sensors were fixed to the elastic bands using both the band's pressure and tape to prevent relative displacement between the sEMG sensors and the muscles during data collection.

3.1.2. Data measurement

A software introduced in a previous research [31] was used to simultaneously collect sEMG signals and finger joint activity data. As shown in Fig. 5, the software visualize the sEMG signals and finger joint motion data, respectively. As shown in Fig. 5, the amplitude information of the sEMG signals and the average Mean Absolute Value (MAV) for all channels were displayed during data measurement. Since MAV is a feature that reflects the average amplitude of the signal [4], the lower right display allows participants to autonomously control the activation intensity of their muscle groups. This ensures that the co-contraction levels of the muscles during different continuous actions

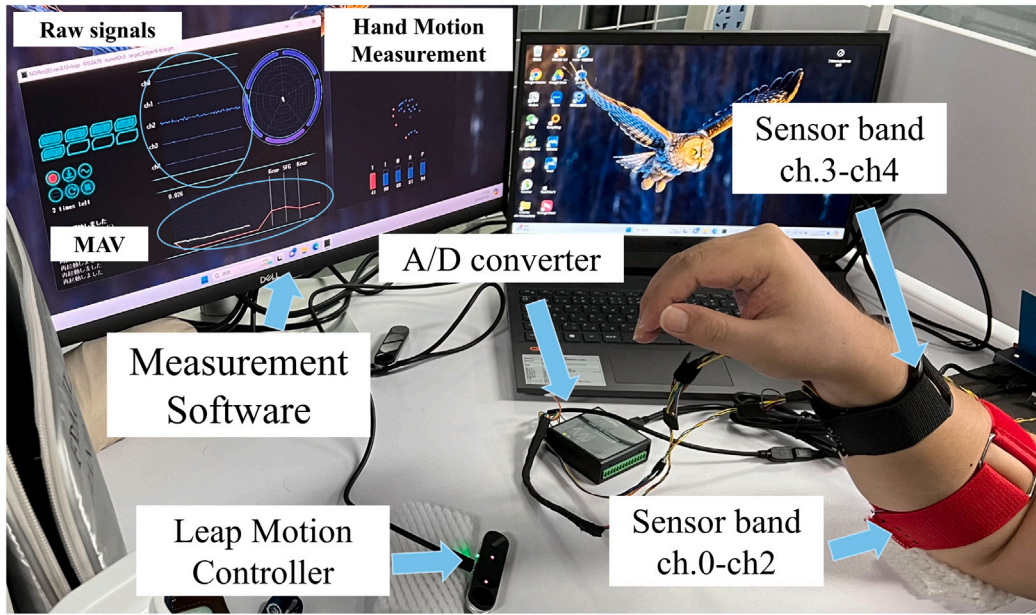


Fig. 5. Data measurement scene.

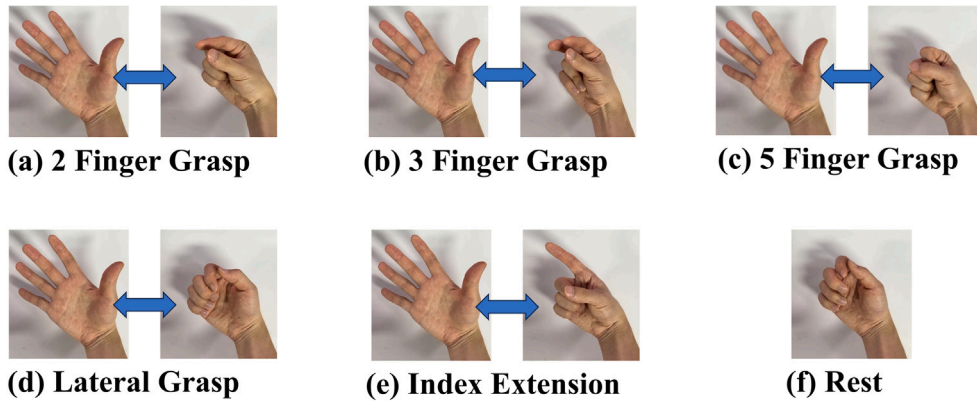


Fig. 6. Postures used in the training data: (a) 2 finger grasp, (b) 3 finger grasp, (c) power grasp, (d) lateral grasp, (e) index finger extension, and (f) rest. The blue bidirectional arrows in the figure indicate that the action includes a transitional process during the motion switch.

were indirectly regulated [10], thereby obtaining more robust sEMG data.

To ensure practicality, we referred to the postural synergy theory of the human hand in prior studies [23] and selected five continuous movement processes and one static posture based on principal components and the actual use of prosthetic hands. Fig. 6 presents all the movements utilized in this study, including the dynamic processes of five movements and one static posture. The five continuously changing target control postures are 2 finger grasp (2FG), 3 finger grasp (3FG), power grasp (PG), lateral grasp (LG), and index finger extension (IE). During the data collection process, participants were instructed to adjust the real-time MAV to closely match the preset MAVs while following the instructions marked on the vertical line to perform a specific action or maintain and switch actions. For example, in Fig. 7, “2FG” indicates that the participant needs to transition from the current posture (Open) to the 2 Finger Grasp posture within the next second. The period from 2FG to the subsequent “Keep” is referred to as the transition process. “Keep” signifies maintaining the current action until the next switching instruction is given. Therefore, when the Open instruction is issued, the participant needs to switch from the 2FG posture to the Open posture.

Five right-handed male subjects (age: 25.8 ± 5.7) participated. During data collection, participants kept their right hand on a horizontal

surface and adjusted it to be clearly captured by the Leap Motion Controller. 12 trials were collected per day over five days, resulting in 60 trials per participant. Every six consecutive trials formed a session, followed by the removal of sEMG sensors and a rest interval of at least three hours. As shown in Fig. 7, in each trial, participants repeated the movements twice, with a five-second rest between movements. After each trial, a five-minute break was given to prevent muscle fatigue. These data were subsequently used to assess the overall performance of the proposed model under various experimental conditions, including within-session, multi-day, and multi-day cross-subject scenarios. Additionally, to investigate the impact of sensor placement variability on model performance, data from seven sessions involving two healthy subjects were collected, with deliberate alterations to sensor positions in each session. The specific experimental conditions and objectives are detailed in the following chapter.

3.2. Performance evaluation of the proposed model

The correlation coefficient (CC), normalized root-mean-square error (NRMSE), and R^2 were selected as the main metrics for model

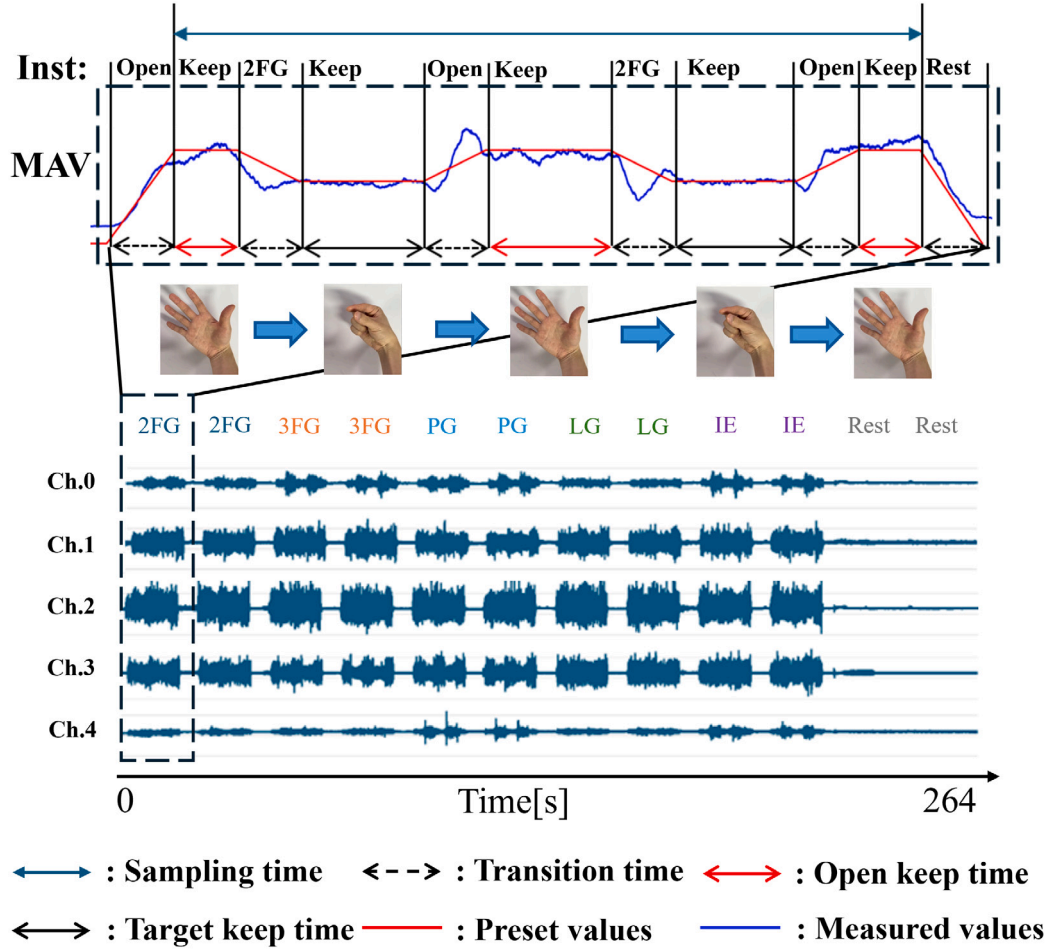


Fig. 7. An example of a trial of sampling comprised of sEMG signals of five continuous movements and a static posture shown in Fig. 6. The dashed box illustrates the indicative MAV variation process of 2 finger grasp. Inst: instructions given to subjects on the actions they need to perform when the measured values reach the corresponding vertical lines. “Sampling time”: the effective sampling duration of the motion sequence, “Transition time”: the period of transitioning from one motion to the next, “Keep time”: the duration for which the posture is maintained.

evaluation. The calculations of the metrics are shown as follows:

$$CC = \frac{\sum_{i=1}^n (Y_i - \bar{Y})(\hat{Y}_i - \bar{\hat{Y}})}{\sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2 \sum_{i=1}^n (\hat{Y}_i - \bar{\hat{Y}})^2}} \quad (5)$$

$$NRMSE = \frac{\sqrt{\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2}}{Y_{max} - Y_{min}} \quad (6)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (Y_i - \hat{Y}_i)^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2} \quad (7)$$

where Y_i and $\hat{Y}_i$ represent the actual and predicted values respectively, $\bar{Y}$ and $\bar{\hat{Y}}$ are their corresponding means, and n is the total number of samples. CC measures the strength and direction of the linear relationship between the predicted and actual values. NRMSE provides a normalized measure of the average magnitude of the errors between predicted and actual values, allowing comparison across different scales. R^2 indicates the proportion of the variance in the dependent variable that is predictable from the independent variables. Additionally, a hand model was created using Blender (Blender Foundation, Amsterdam) to simulate the postures that the control algorithm would generate on a myoelectric prosthetic hand based on the predicted continuous joint angle changes from the sEMG data in the test set.

3.3. Hyperparameter tuning

To address the hyperparameter selection process, we detail the settings and tuning methodology used in this study. The hyperparameters

were selected based on empirical experiments and adjusted to optimize model performance. Specifically, the learning rate was initialized at 0.001 and adjusted dynamically using a StepLR scheduler, which reduced the learning rate by a factor of 0.5 every 40 epochs (10% of the total epochs). The batch size was set to 256, ensuring efficient training while maintaining memory feasibility. The model was trained for 400 epochs, with early stopping implemented to prevent overfitting. The optimizer used was Adam [34], chosen for its robustness and ability to handle sparse gradients, which is critical for deep learning tasks. Additionally, Kaiming initialization [35] was applied to the model weights to facilitate effective training by maintaining consistent variance across layers.

These hyperparameter choices were informed by experimental observations, and the learning rate and batch size were fine-tuned through iterative testing rather than formal cross-validation due to resource constraints. This configuration demonstrated stable convergence and robust performance across the training and validation datasets.

4. Results and discussion

4.1. Results of within-session experiment

In this experiment, five subjects were trained independently. For each subject, the training set and the testing set were from the same session. In a same session of the data, the sensor bands were worn only once and would not be removed during the data measurement.

Table 2
Average statistics for all subjects of within-session experiment.

Motion	Metrics	Joint Angle					
		Thumb Extension	Thumb Rotation	Index	Middle	Ring	Pinkie
2FG	CC	–	0.963 ± 0.030	0.959 ± 0.033	0.973 ± 0.028	0.979 ± 0.024	0.974 ± 0.029
	R2	–	0.902 ± 0.090	0.890 ± 0.081	0.922 ± 0.087	0.947 ± 0.057	0.939 ± 0.070
	NRMSE	0.074 ± 0.118	0.097 ± 0.043	0.082 ± 0.036	0.102 ± 0.052	0.082 ± 0.039	0.084 ± 0.041
3FG	CC	–	0.964 ± 0.043	0.952 ± 0.058	0.954 ± 0.052	0.977 ± 0.035	0.969 ± 0.054
	R2	–	0.870 ± 0.192	0.882 ± 0.133	0.809 ± 0.196	0.935 ± 0.101	0.902 ± 0.186
	NRMSE	0.068 ± 0.098	0.105 ± 0.086	0.077 ± 0.043	0.096 ± 0.049	0.086 ± 0.061	0.100 ± 0.083
PG	CC	–	0.974 ± 0.020	0.969 ± 0.025	0.987 ± 0.010	0.988 ± 0.010	0.986 ± 0.011
	R2	–	0.929 ± 0.058	0.906 ± 0.091	0.968 ± 0.026	0.971 ± 0.022	0.968 ± 0.024
	NRMSE	0.053 ± 0.047	0.088 ± 0.029	0.118 ± 0.055	0.071 ± 0.029	0.068 ± 0.027	0.072 ± 0.026
LG	CC	0.850 ± 0.171	0.921 ± 0.049	0.942 ± 0.057	0.952 ± 0.047	0.948 ± 0.048	0.937 ± 0.053
	R2	0.550 ± 0.605	0.710 ± 0.314	0.840 ± 0.167	0.885 ± 0.099	0.879 ± 0.095	0.852 ± 0.117
	NRMSE	0.221 ± 0.129	0.086 ± 0.037	0.121 ± 0.065	0.094 ± 0.038	0.087 ± 0.031	0.093 ± 0.036
IE	CC	–	0.947 ± 0.095	–	0.955 ± 0.087	0.958 ± 0.089	0.957 ± 0.081
	R2	–	0.878 ± 0.191	–	0.886 ± 0.235	0.907 ± 0.180	0.904 ± 0.175
	NRMSE	0.065 ± 0.086	0.096 ± 0.064	0.137 ± 0.078	0.095 ± 0.071	0.087 ± 0.064	0.087 ± 0.060
Rest	NRMSE	0.032 ± 0.021	0.017 ± 0.013	0.014 ± 0.008	0.013 ± 0.011	0.012 ± 0.011	0.011 ± 0.011
All	CC	0.898 ± 0.116	0.949 ± 0.048	0.936 ± 0.047	0.962 ± 0.032	0.971 ± 0.029	0.965 ± 0.041
	R2	0.792 ± 0.257	0.898 ± 0.095	0.872 ± 0.090	0.921 ± 0.078	0.940 ± 0.059	0.928 ± 0.082
	NRMSE	0.117 ± 0.077	0.092 ± 0.041	0.106 ± 0.040	0.089 ± 0.037	0.080 ± 0.035	0.085 ± 0.040

This ensured that the difference of the sEMG signals mainly came from the muscle fatigue. Due to overlapping segments in the input signals, the redundant joint angle predictions are removed during the actual prediction process. To smooth the predicted angles, a moving average filter with a window length of 50 is applied to the prediction results.

Table 2 presents the performance of the angle predictor trained under the given experimental conditions for various movements. The results in the table are based on the mean and standard deviation of all sessions across all subjects. For the thumb extension angle, the CC and R² values were only calculated for the LG movement, where noticeable changes were observed, and not for other movements. Similarly, for the MCP angle of the index finger, CC and R² values were not calculated for the IE movement due to the lack of significant changes. The evaluation results for all movements are based on the complete dataset of the corresponding test sets. Bold values indicate the top two scores for each metric across the joint angles.

For the five angles ranging from thumb rotation to pinkie, the highest CC and R² scores were achieved during the PG movement, with the second-highest scores primarily observed during the 2FG movement. Notably, although the second-highest CC score for thumb rotation appeared in the 3FG movement, the score for the 2FG movement was very close, with a difference of only 0.001 in the mean value. This suggests that the sEMG features in the PG and 2FG movements are more representative compared to other movements, making it easier to obtain favorable results with these inputs. For the thumb extension angle, the CC and R² values were 0.850 ± 0.171 and 0.550 ± 0.605, respectively, which were lower than the highest scores for the other angles. This discrepancy might be attributed to the fact that the angle changes occur only in the LG movement, leading to an imbalance in the training data compared to the other angles.

For all angles, the lowest NRMSE scores were observed in the Rest state, while the second-lowest NRMSE scores showed greater variability compared to the CC and R² scores. For the first three angles, the second-lowest NRMSE scores appeared in the IE, LG, and 3FG movements, respectively, while for the remaining angles, they occurred in the PG movement. This phenomenon may be due to the fact that joint angles in the Rest state are typically fixed, and the sEMG signals in this state are highly distinguishable, resulting in higher prediction accuracy. However, during dynamic movements, even if the brain's commands (e.g., moving from an open-hand posture to the 2FG target posture) are the same, the target joint angles may not be identical. Moreover, measurement errors from the Leap Motion device cannot be ignored,

as hand postures tend to vary slightly when facing the camera from different angles.

Fig. 8 shows the prediction results of the trained TF2AngleNet for a set of movements in the test set. The data used in Fig. 8 is from the first trial of Subject 5. The red line represents the measured data, while the blue line represents the predicted data after applying a moving average filter. Fig. 9 illustrates the actual control effects of the predicted signals on each hand movement of the model. The data used in Fig. 9 were obtained from 12 evenly spaced points within the control signal. The comparisons of different metrics corresponding to various actions were presented using heatmaps, and the results were generally consistent with those shown in Table 2.

4.2. Performance comparison between proposed and single-domain models

In this section, this paper compares the proposed method using data from the Within-Session condition with networks that utilize single-domain features only. The single-domain networks differ from the network structure shown in Fig. 3 by having a single-channel feature extraction module and lacking the feature concatenation part, while all other parameters remain the same. To ensure that the classifiers receive the same amount of features as the proposed network structure, the feature modules of the single-channel networks were adjusted accordingly, maintaining the same number of features for decision-making. Additionally, the parameters of the three networks were kept as similar as possible to avoid performance improvements due to increased model complexity.

Networks utilizing only time-domain and frequency-domain features are abbreviated as RawOnlyCNN and FreqOnlyCNN, respectively. Their inputs were 1D raw time-domain signals and 2D time–frequency feature maps and their number of input features were the same. The loss variations under different models for all sessions of the five subjects are shown in Fig. 10, with the Mean Absolute Error (MAE) used as the loss function. TF2AngleNet performed best, with the final test loss decreasing to 0.065 and a generalization gap of 0.029. RawOnlyCNN performed the worst, with a final loss of 0.091 and a generalization gap of 0.045. The performance of FreqOnlyCNN was close to that of TF2AngleNet, with a final loss of 0.071 and a generalization gap of 0.033. Each model was saved at the point of minimum loss, and the corresponding metrics on the test set were then calculated. Table 3 shows the average values of various metrics for different models across

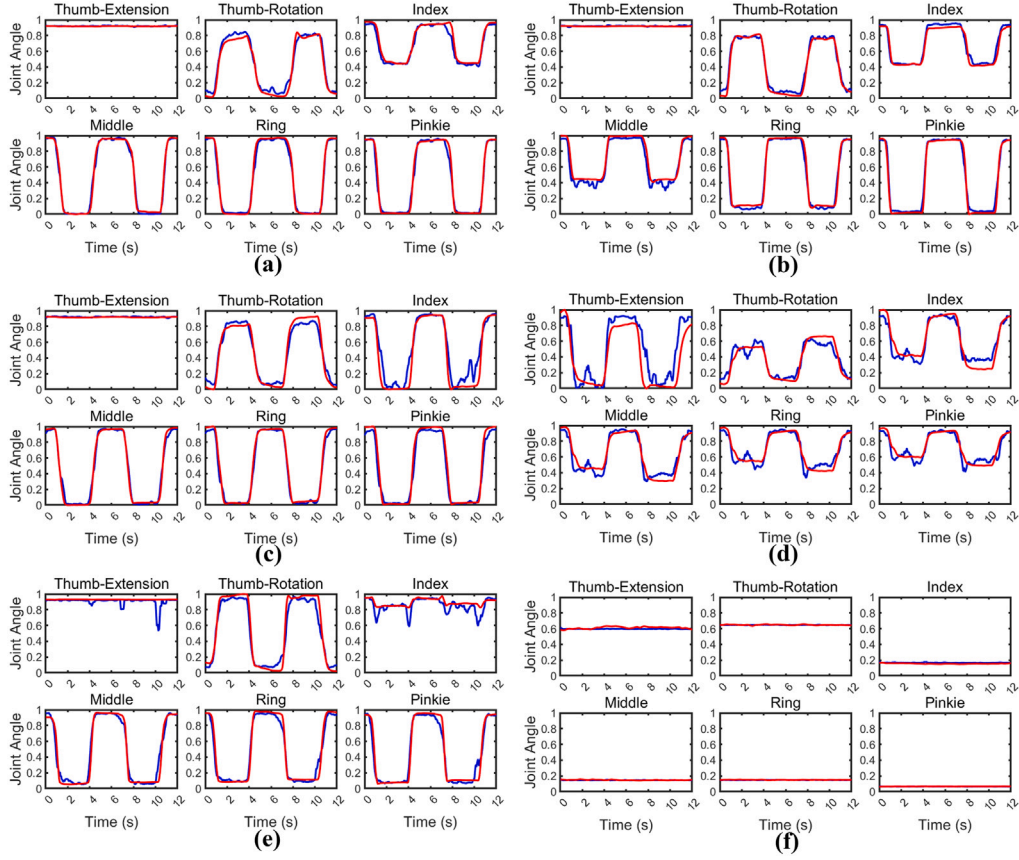


Fig. 8. An example of joint angle prediction using trained TF2AngleNet. The red line represents the measured data, and the blue line represents the predicted data after applying a moving average filter. (a) 2 finger grasp, (b) 3 finger grasp, (c) power grasp, (d) lateral grasp, (e) index finger extension, and (f) rest.

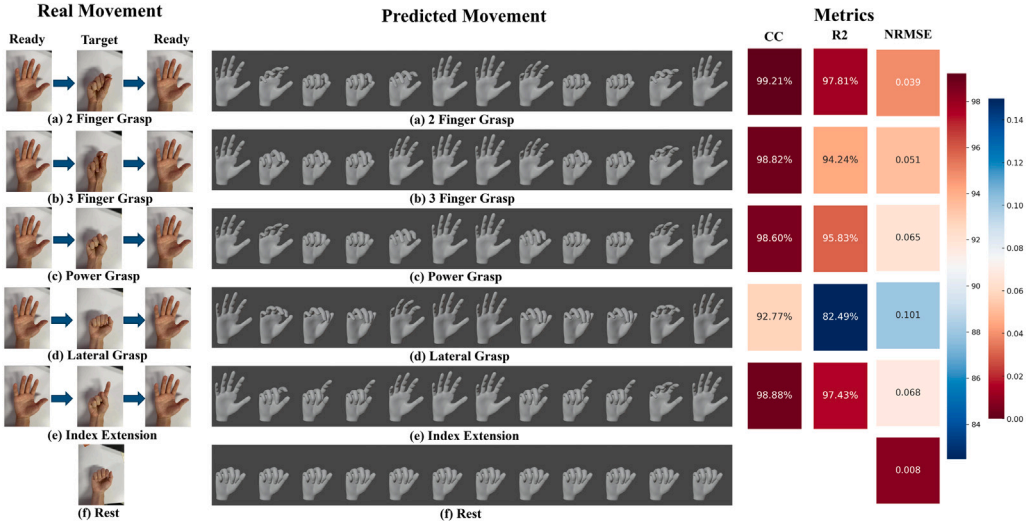


Fig. 9. Visualization of an example of joint angle prediction using a virtual hand model by the trained TF2AngleNet. (a) 2 finger grasp, (b) 3 finger grasp, (c) power grasp, (d) lateral grasp, (e) index finger extension, and (f) rest.

all joint angles, with the best and second-best values highlighted in dark gray and light gray, respectively. Compared to the other two models, TF2AngleNet achieved average improvements of 1.23%, 2.42%, and 12.43% in CC, R^2 , and NRMSE, respectively. Furthermore, paired t-tests comparing TF2AngleNet with RawOnlyCNN ($CC:p = 8.69E-05 < 0.01$, $R^2:p = 8.38E-06 < 0.01$, $NRMSE:p = 9.34E-12 < 0.01$) and FreqOnlyCNN ($CC:p = 1.12E-02 < 0.05$, $R^2:p = 3.54E-02 < 0.05$,

$NRMSE:p = 3.40E-05 < 0.01$) revealed that these improvements were statistically significant. These results indicate that the two-dimensional time–frequency spectrogram contains more effective information for decoding finger movements than the raw sEMG signals. By combining the time-domain and frequency-domain features of EMG signals under the same movement conditions, the proposed method enhances the decoding performance of finger movements based on EMG signals.

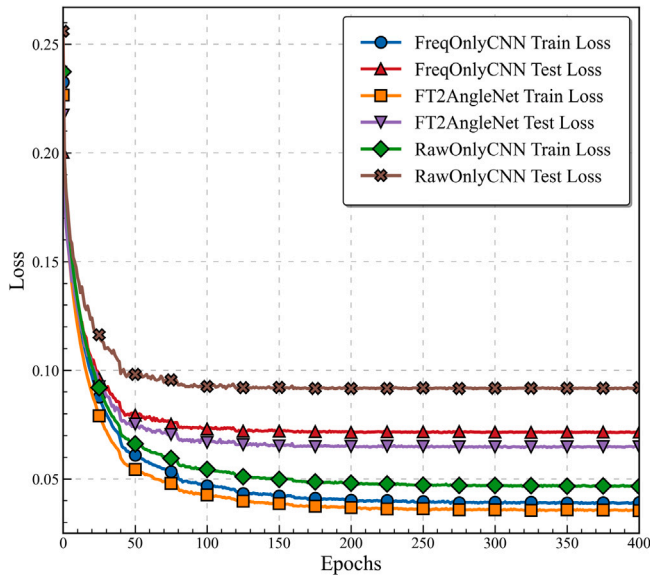


Fig. 10. Loss curves of three models.

Table 3
Comparison with single-domain models.

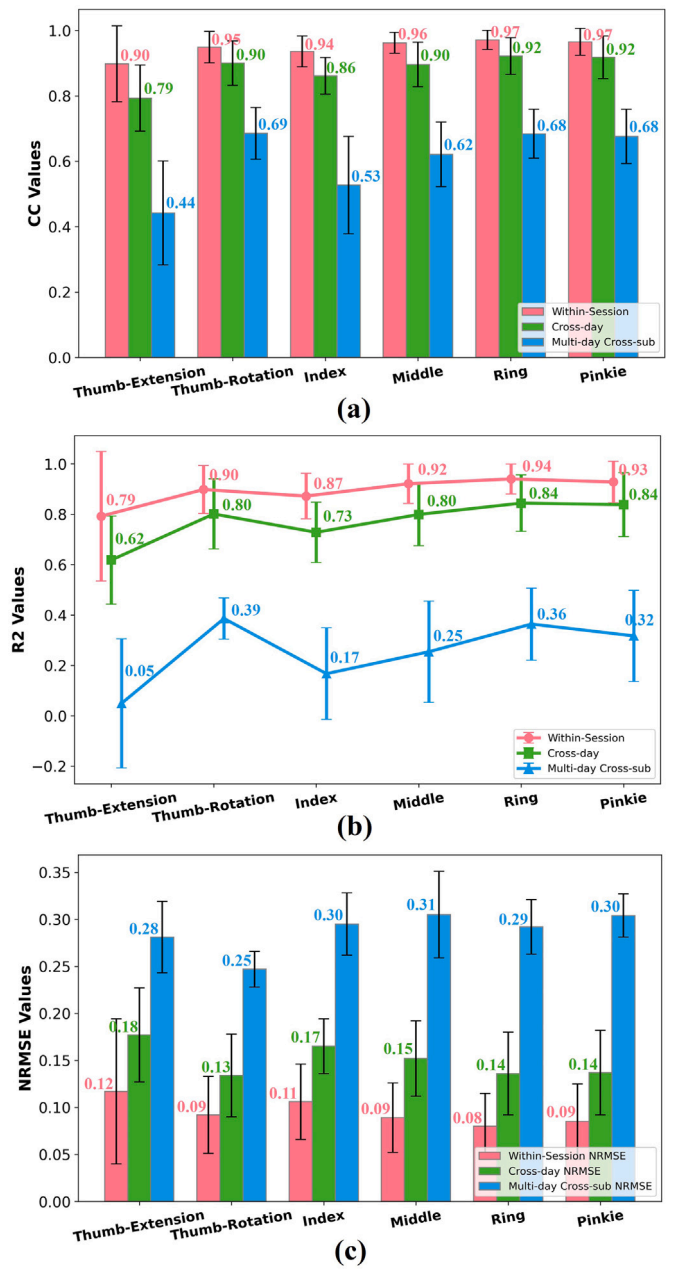
Model	Metrics		
	CC	R2	NRMSE
TF2AngleNet	0.947	0.892	0.095
RawOnlyCNN	0.934	0.866	0.110
FreqOnlyCNN	0.937	0.876	0.107

4.3. Comparison results of within-session experiment, multi-day experiment and multi-day cross-subject experiment

This study is positioned in the first stage of the proposed conceptual framework, which involves collecting a large amount of EMG signals from the general people to extract common signal features. In this section, 2 experimental conditions (Multi-day and Multi-day Cross-subject) were designed to evaluate the impact of sEMG time-dependent factors (such as muscle fatigue, sweating, and electrode displacement) and individual differences on model performance, thereby demonstrating the validity of the framework and highlighting the necessity for the second stage.

Cross-Day experiment validates the generalization ability of the proposed method under long-term use by testing on unseen days for each subject. For the Cross-Day experiment, a leave-one-day-out cross-validation approach was adopted, resulting in a five-fold cross-validation process, as the dataset consists of five days of data. For each fold, data from four days were used as the training set, and data from the remaining day were used as the test set. The Multi-day Cross-subject experiment evaluates the model's generalization performance on unseen subjects over extended time spans. For this experiment, leave-one-subject-out cross-validation was employed, where each subject's data was used as the test set while the remaining four subjects' data formed the training set.

The reported results represent the average performance across the five folds in both experiments, aiming to provide a rigorous evaluation of the model's generalization ability under unseen conditions. The average performance of the model in terms of CC, R^2 , and NRMSE for unseen days and unseen subjects is presented in Fig. 11. The specific values of the metrics depicted in the figure are the mean values rounded to two significant digits, illustrating the model's performance under different conditions.

Fig. 11. CC, R^2 , NRMSE values across different experiments.

The comparison of the metrics for the three experiments is illustrated in Fig. 11. The metrics of all three experiments were analyzed using ANOVA, and the p-values were all less than 0.001. The CC decreased by an average of 6.94% from within-session to Cross-Day, and further decreased by 31.63% from Cross-Day to Multi-day Cross-subject conditions. The R^2 score showed a reduction of 13.74% on average from within-session to Cross-Day, followed by an additional decrease of 68.03% from Cross-Day to Multi-day Cross-subject conditions. The NRMSE increased by an average of 36.93% from within-session to Cross-Day, with a further increase of 47.56% from Cross-Day to Multi-day Cross-subject conditions.

These results indicate that the model demonstrates greater overall robustness under Cross-Day scenarios compared to Multi-day Cross-subject conditions, highlighting that domain shifts between subjects

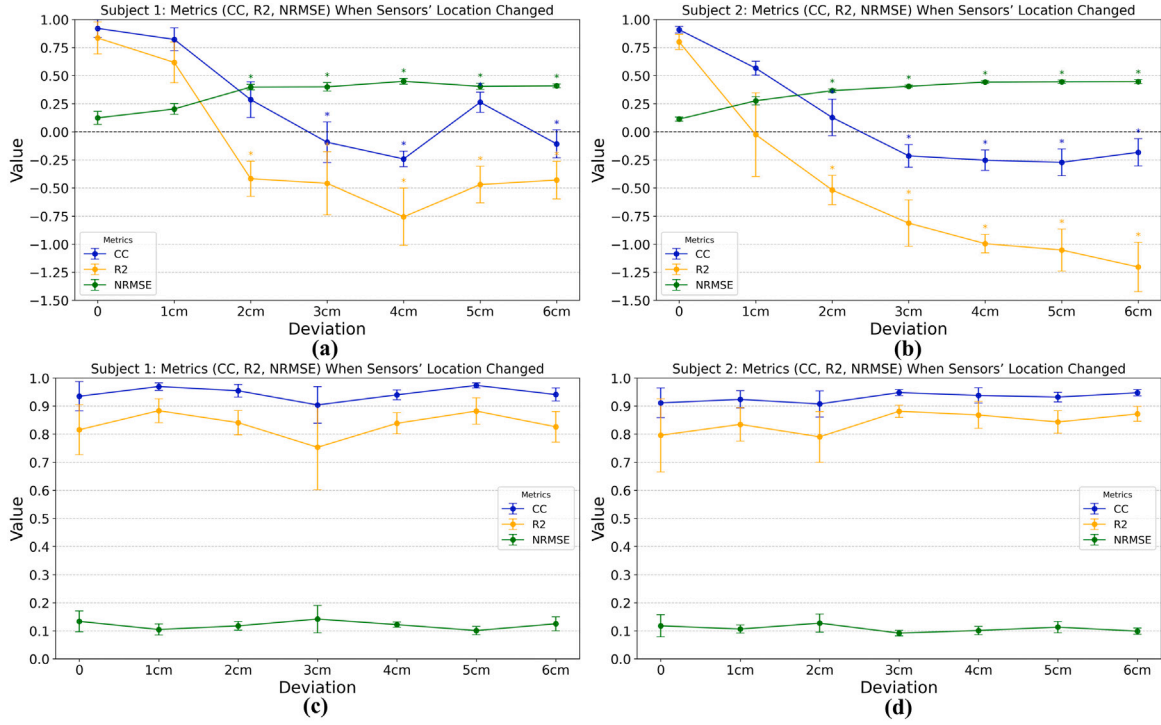


Fig. 12. Impact of sensor placement variability on model performance. (a), (b): Predictive performance of TF2AngleNet for Subject 1 and Subject 2 under varying sensor displacement conditions when the model was trained using data from data with correct sensor placement only. (c), (d): Predictive performance of TF2AngleNet for Subject 1 and Subject 2 under varying sensor displacement conditions when the model was trained using data from sessions with multiple sensor placements.

have a more substantial impact on model performance than temporal factors alone. The findings highlight the critical need to address inter-subject variability to improve model robustness and reliability.

4.4. Impact of sensor placement variability on model performance

To systematically evaluate the impact of sensor placement variability to the model performance, two experiments were conducted. Data from seven sessions involving two healthy subjects were collected, where sensor positions were deliberately altered in each session. The experimental setup was designed to simulate potential sensor shifts that prosthesis users might encounter in daily life. Specifically, the displacement conditions were defined by shifting the two sensor bands simultaneously along the skin surface in a clockwise direction by specific distances (measured in centimeters) from the optimal placement, as shown in Fig. 4.

Experiment 1: This experiment aims to investigate the predictive performance of the network on sEMG signals under different conditions when the model was trained using data with correct sensor placement only. Every session was divided into three parts which contains two trials of data, and every part was used as test set once to form a three-fold cross validation process. For each subject, TF2AngleNet was trained using data from the session with correct sensor placement, as described in the manuscript. The Fig. 12(a), (b) illustrate the model's performance metrics under varying degrees of sensor displacement for Subject 1 and Subject 2, respectively. The results indicate that the model maintained a relatively low level of predictive performance with a 1 cm displacement. However, when the displacement increased to 2 cm or more, the model's predictive ability became relatively poor ($p < 0.05$). This decline is likely due to significant changes in the recorded muscle activation patterns, which the model had not encountered during training.

Experiment 2: This experiment aims to investigate the predictive performance of the network on sEMG signals under different conditions when the training data includes sEMG data from multiple sensor

placements. Similar to experiment 1, each session was divided into three parts, each containing two trials of data. Each part was used as a test set once in a three-fold cross-validation process. For each subject, TF2AngleNet was trained using data from seven sessions with different sensor placements, as described above. As shown in Fig. 12(c), (d), TF2AngleNet maintained relatively high prediction performance across different EMG conditions after learning from diverse scenarios. The average values of the CC, R^2 , NRMSE metrics for the models of the two subjects under different conditions were 0.938, 0.838, and 0.114, respectively. Compared to the performance under within-session conditions reported in the manuscript, these values showed a decrease of 0.95%, 6.04%, and 20.00%, respectively. These results indicate that the model demonstrates strong generalization ability across varying sensor placement conditions, making it robust to practical variability in sensor positions.

4.5. Discussion on the feature importance of the proposed model

To enhance the interpretability of the proposed network, a feature importance analysis was conducted. Feature importance is quantified by measuring the performance change of the model when each feature channel is zeroed out. Specifically, the contribution of each feature channel to the model's performance across different metrics (CC, R^2 , and NRMSE) was analyzed under the within-session experimental condition for each subject. For this analysis, we iteratively zeroed out each feature channel in the spectrogram and raw sEMG data and assessed the absolute impact on the model's performance. To ensure generalizability, we averaged the feature importance scores across all subjects. For a given feature channel c , its importance I_c is calculated as:

$$I_c = |P_{\text{original}} - P_{\text{zeroed},c}| \quad (8)$$

where I_c represents the importance of feature channel c , P_{original} denotes the model's performance with the complete input data, and $P_{\text{zeroed},c}$ denotes the model's performance when channel c is zeroed out in the input data.

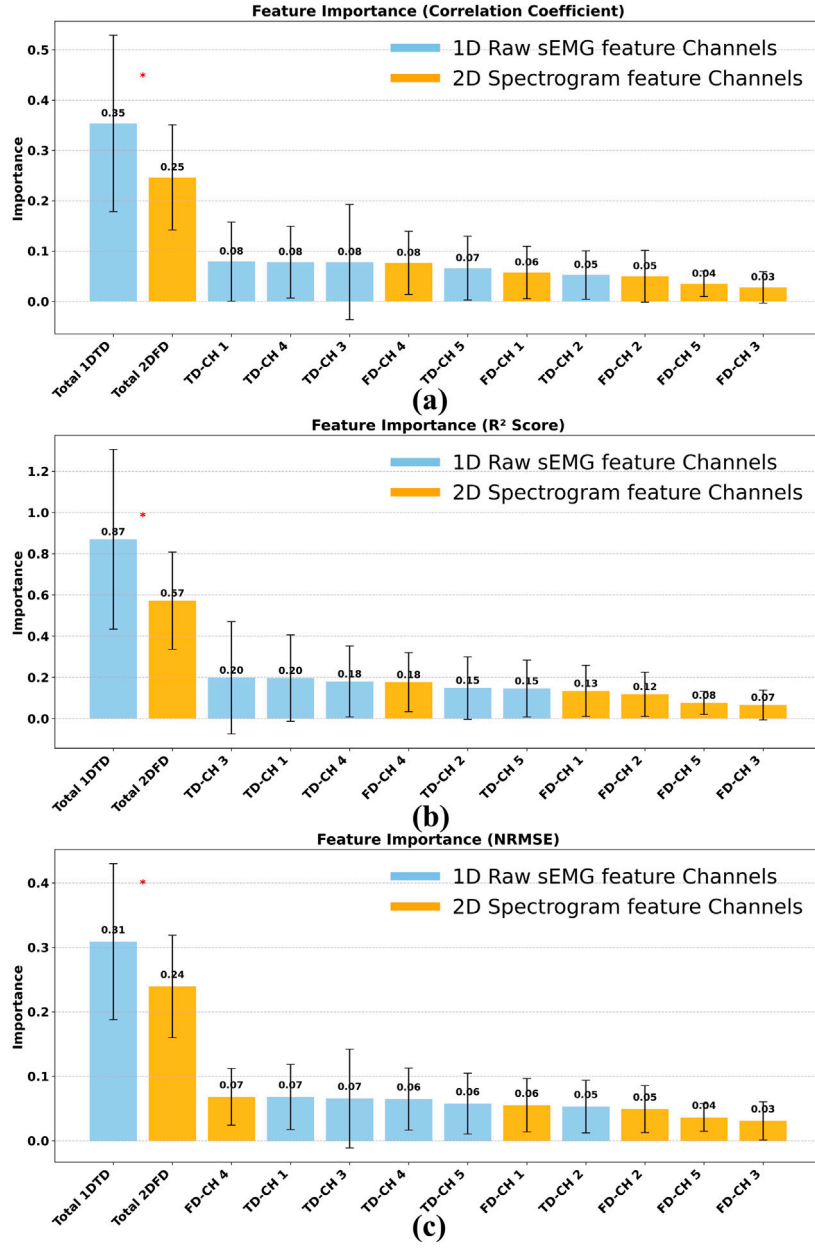


Fig. 13. Feature importance analysis. TD-CH x denotes 1D raw sEMG features from channel x , FD-CH x denotes 2D spectrogram features from channel x , and Total 1D/2D is the sum of importance values across all channels for each feature type.

To evaluate the overall contribution of a feature type (e.g., raw sEMG data or spectrogram data), the total feature importance is defined as the sum of the importance values of all channels of a feature type. The results are presented in Fig. 13, which shows the feature importance of spectrogram and raw sEMG channels and their total contributions to CC, R^2 , and NRMSE. Our findings indicate that the contribution of raw sEMG features surpasses that of spectrogram features by 40.0%, 52.6%, and 29.2% for the CC, R^2 , and NRMSE metrics, respectively. However, when examining individual feature channels, we observed that while raw sEMG channels occupy the top 6 positions in feature importance rankings, the fourth spectrogram channel consistently ranks among the top 5 in all metrics and is the top contributor for NRMSE. These results demonstrate that both feature types play complementary roles in predicting finger joint angles, thereby enhancing the reliability and trustworthiness of the model.

4.6. Discussion on the implementation of the proposed approach

Considering the feasibility of implementing this method, the time complexity and the computation time of the proposed model was calculated. For single input data comprised of both spectrograms and raw sEMG signals, the proposed TF2AngleNet's computational complexity is 18.06M Flops. Using an NVIDIA RTX 3070 Ti Laptop GPU, the average inference time of the TF2AngleNet model was measured over 100,000 trials. The results indicate that the model achieves an average inference time of approximately 2040 μ s per sample, corresponding to a processing speed of 490 samples per second. Considering that only four input data points are required per second (based on a spectrogram shift size of 500 and a sampling frequency of 1600 Hz), the model demonstrates computational efficiency and satisfies the real-time requirements for prosthetic hand control. While this study utilized a high-performance GPU for evaluation, the model's lightweight architecture suggests potential compatibility with embedded systems. The proposed model has

Table 4
Comparison with Previous Studies.

Research	Angles	Subjects	Sensors	Method	Results		
					CC	R2	NRMSE
[24]	3	10	32	CNN	0.950	–	–
[19]	5	9	32	Synergy-MSM	0.931	–	0.107
[15]	22	25	10	LSTA-Conv	0.660	0.316	–
	22	35	12	LSTA-Conv	0.825	0.727	–
	18	16	12	LSTA-Conv	0.875	0.890	–
This	6	5	5	TF2AngleNet	0.947	0.892	0.095

a total of 104,942 trainable parameters, and there is still room for optimization.

In this study, the collection of sEMG signals and hand movement data from participants was performed on the same side of the body. For amputees, it is planned to instruct them to issue identical movement commands to both sides, allowing us to use the hand movement data from the intact side as labels for the sEMG data of the affected side. The effectiveness of this experimental design was proved in some previous studies [36,37].

4.7. Comparison with previous studies

Table 4 presents a comparison of recent studies on predicting hand joint angles based on sEMG signals with the current study. Qin et al. [24] introduced a CNN-based regression model for simultaneously predicting three forearm movements — wrist flexion/extension, arm pronation/supination, and hand gripping/opening — achieving the highest CCs of 97%, 96%, and 92%, respectively, with the CC value in the Table 4 representing the average of the highest CCs across the three DOF. He et al. [19] developed a synergy-musculoskeletal model (Synergy-MSM) to predict five angles of the wrist, thumb, index finger, and middle finger. The CC represented their average value of CCs from all angles. Long et al. used data of six motions from three public datasets to evaluate their LSTA-Conv neural network and the metrics were obtained directly from their results. Based on the results, the current study achieved comparable or better accuracy in predicting multi-joint angles using the fewest number of sensors compared to previous studies [15,19,24].

Additionally, feature selection and the hand data measurement method are key highlights of this study. Previous studies predominantly utilized time-domain features of EMG signals, whereas this study explored the combination of 1D time-domain features and 2D time-frequency maps rich in frequency-domain information for predicting joint angles. The non-contact hand data measurement approach employed in this study simplifies the data acquisition process, reduces the barrier for users to implement this algorithm, and avoids the discomfort associated with direct contact between the hand and sensors.

4.8. Limitations and future work

This study has several limitations, which will be addressed in future work:

Controlled Environment Testing: The control performance of the proposed method was only validated through a virtual hand model in a controlled laboratory environment. This limits the applicability of the findings to real-world scenarios. To address this, we plan to test the method on a multi-DOF myoelectric prosthetic hand in real-world daily-life scenarios. Experiments will involve diverse environments to further validate the method's effectiveness and bionicity.

Sensor Placement Variability: Although adjustments such as external palpation were performed to minimize placement variability during data collection, this manual approach may not completely eliminate variability. The study did not include advanced sensor calibration or real-time adaptive mechanisms to compensate for potential sensor

misalignment. Future work will address this limitation by exploring advanced calibration techniques and developing adaptive algorithms capable of real-time compensation for sensor misplacement. Additionally, data augmentation strategies during training will be considered to enhance the model's robustness to sensor placement variability, ensuring consistent performance in real-world scenarios.

Limited Participant Diversity: The study only included a small sample of healthy, right-handed male participants, which may impact the generalizability of the findings. Additionally, the exclusion of female, left-handed, and elderly participants limits the diversity of the dataset. Future studies will expand the dataset by including participants with diverse demographics, such as female, left-handed, and elderly individuals, to enhance the model's robustness and applicability to a broader population.

Applicability to Amputee Users: Due to the lack of muscle usage for controlling different fingers, there are often significant differences in sEMG signals between individuals with upper limb deficiencies and healthy individuals. This study presents the first stage of the proposed conceptual framework and does not directly address these differences. Therefore, for the second stage which involves using transfer learning approaches, in future work, a method that maps the sEMG signals of amputee users to those of healthy individuals will be developed. This method is inspired by CycleGAN [38], a generative adversarial network designed for style transfer in images. This approach will ensure the method's adaptability to individuals with upper limb deficiencies and enhance its practical relevance.

Performance on Resource-Constrained Devices: The inference performance of the proposed model was validated on a high-performance GPU. While the results demonstrate computational efficiency, this setup does not fully represent the constraints of low-power embedded systems used in prosthetic devices. To address this, we plan to explore model optimization techniques, such as quantization and pruning, to enhance compatibility with resource-constrained environments. Future work will include testing the optimized model on embedded platforms to ensure feasibility for real-world applications.

5. Conclusion

This study developed a TF2AngleNet, which leverages time-frequency domain information from sEMG signals for simultaneous regression of multiple finger joint angles. The results of six-motion and six-joint-angle regression demonstrated that the proposed approach achieves high performance in within-session conditions. Evaluation metrics and visualized outcomes showed that the method effectively and naturally predicted multiple joint angles simultaneously. This research provides a promising solution for flexible, biomimetic, and practical myoelectric hand control methods.

CRedit authorship contribution statement

Hai Jiang: Writing – original draft, Software, Methodology, Investigation, Conceptualization. **Yusuke Yamanoi:** Writing – review & editing, Conceptualization. **Peiji Chen:** Writing – review & editing, Methodology, Investigation. **Xin Wang:** Writing – review & editing.

Shixiong Chen: Writing – review & editing. **Xu Yong:** Writing – review & editing, Resources. **Guanglin Li:** Writing – review & editing, Supervision, Resources. **Hiroshi Yokoi:** Writing – review & editing, Supervision, Resources. **Xiaobei Jing:** Writing – review & editing, Resources, Project administration.

Ethical approval

All experimental procedures was approved by the Ethics Committee of The University of Electro-Communications, Tokyo, Japan (H10006(7)).

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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